Algorithm

1. **Load Dataset**
    a. Read the dataset from its source (CSV, Excel, or PDF).
    b. Ensure that all files are accessible and readable.
2. **Inspect Dataset**
    a. Check the number of rows and columns.
    b. Identify column names, data types, and basic statistics.
    c. Note any missing or inconsistent values.
3. **Clean Dataset**
    a. Handle missing values (drop rows, fill with mean/median/mode, or use imputation).
    b. Remove duplicate rows if present.
    c. Correct any inconsistent formatting (dates, categories, etc.).
4. **Preprocess Data**
    a. Encode categorical variables (e.g., one-hot or label encoding).
    b. Normalize or standardize numeric features.
    c. Split dataset into training set, validation set, and test set.
5. **Feature Selection / Extraction**
    a. Identify important features based on correlation, variance, or domain knowledge.
    b. Optionally, reduce dimensionality using PCA or other techniques.
6. **Model Selection**
    a. Choose a suitable algorithm (e.g., regression, classification, clustering) based on the task.
    b. Initialize model parameters.
7. **Train Model**
    a. Feed the training data to the model.
    b. Adjust hyperparameters if needed for better performance.
8. **Validate Model**
    a. Evaluate the model on the validation set.
    b. Record accuracy, precision, recall, F1-score, or relevant metrics.
    c. Fine-tune hyperparameters if performance is low.
9. **Test Model**
    a. Evaluate the final model on the test set.
    b. Ensure the performance is consistent with validation results.

10. **Deploy / Predict**
    a. Build a function or interface that takes new data and outputs predictions.
    b. Optionally log predictions with confidence scores.
11. **Generate Dataset PDF (Optional)**
    a. Create a PDF version of the dataset with sample rows, column descriptions, and summary statistics.
    b. Include visualizations if possible, for clarity.
12. **Document Results**
    a. Summarize data insights, model performance, and key observations.
    b. Include methodology and workflow for evaluators to follow

Dataset Sample (CIFAR-100 from Kaggle)

Dataset Description:

- **Project:** Coloring black-and-white images
- **Dataset Source:** CIFAR-100 on Kaggle (CIFAR-100 Python)
- **Number of Images:** 60,000 (50,000 training, 10,000 test)
- **Preprocessing:**
    - Converted color images to grayscale (1 channel) for model input
    - Original color images (3 channels) used as ground truth
    - Normalized pixel values to 0–1 for model training

Grayscale Input	Ground Truth Color